

Grade Decomposition of the Lattice QCD Pion Correlator in the Cl(6) Framework

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Abstract

The Clifford algebra $\text{Cl}(6)$ embeds the $\text{SU}(3)$ color gauge group as a subgroup of $\text{Spin}(6) \cong \text{SU}(4)$, making the $\text{SU}(3)$ representation structure of the Wilson–Dirac spinor space algebraically explicit through the decomposition $\mathbf{8} = \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1} \oplus \mathbf{1}$. We construct the corresponding four-channel decomposition of the lattice pion correlator and report its first numerical implementation on quenched gauge ensembles.

The four representation projectors are constructed as Lagrange spectral projectors of the Cartan hypercharge $H = B_{12} + B_{34} + B_{56}$ of $\text{spin}(6)$, whose eigenspectrum $\{\pm\frac{1}{2}, \pm\frac{3}{2}\}$ matches the $\text{SU}(3)$ representation dimensions. They satisfy completeness, idempotency, mutual orthogonality, $\text{SU}(3)$ covariance, charge-conjugation pairing, and commutativity with the 4D chiral operator γ_5 to numerical residual $\leq 2.4 \times 10^{-15}$. The labeling (Γ_7, H) , where $\Gamma_7 = iI_6$ is the $\text{Spin}(6)$ chirality, is the standard $\text{SU}(4) \supset \text{SU}(3) \times \text{U}(1)$ Levi decomposition of the Dirac spinor of $\text{Spin}(6)$, known from representation theory and from the Pati–Salam $\text{SU}(4)_{\text{color}}$ model.

A kinematic limit theorem follows from the algebraic structure: in the free Wilson–Dirac theory, the grade fractions $f_r(t) \equiv \langle C_\pi^{(r)}(t) \rangle / \langle C_\pi(t) \rangle$ equal $\dim(r)/8$ exactly, a consequence of the residual $I_4 \otimes \text{U}(2)$ symmetry of the free operator combined with $\text{SU}(3)$ color and spatial-cubic invariance.

On thermalized quenched ensembles at $N = 6$, $\beta \in \{4.5, 6.0\}$, and bare quark mass $m_0 = 0.05$, the algebraic completeness identity $\sum_r C_\pi^{(r)}(t) = C_\pi(t)$ is verified per configuration to maximum residual 5.4×10^{-16} , and the $\text{Spin}(6)$ -chirality fractions are fixed at $f_{(\Gamma_7=\pm 1)} = \frac{1}{2}$ exactly on every configuration by an identity independent of the gauge dynamics. The measured grade fractions are consistent with the kinematic free-field values at the 10^{-4} level, and the phase differences between confined and deconfined regimes are consistent with zero at heavy quark mass. The decomposition admits no natural analog within the standard four-component Wilson–Dirac algebra: the bivector B_{56} requires the internal generators e_5, e_6 that are absent from $\text{Cl}(4)$, and the resulting grade structure cuts across the γ_5 chirality decomposition. The chiral-regime extension at $\kappa \rightarrow \kappa_c^{\text{ren}}$, required to test whether non-perturbative gauge dynamics

distinguish the four representation channels, is identified as the target for follow-up investigation.

Keywords: Clifford algebra, lattice QCD, Wilson fermions, pion correlator, grade decomposition, Cartan hypercharge, SU(3) representation theory, kinematic limit theorem, quenched approximation.

1 Introduction

1.1 Motivation

The Clifford algebra $\text{Cl}(6)$ provides a natural home for the internal algebraic structure of Wilson lattice QCD. Its eight-dimensional minimal left ideal decomposes under $\text{SU}(3)$ as $\mathbf{3} \oplus \mathbf{\bar{3}} \oplus \mathbf{1} \oplus \mathbf{1}$ —accommodating quark color triplets, antiquark anti-triplets, and two gauge singlets—while the 15 bivectors span the Lie algebra of $\text{Spin}(6) \cong \text{SU}(4)$, which contains $\text{SU}(3)$ as the stabilizer subgroup of a preferred spinor direction. This algebraic architecture was established by Furey [20–24] and is reviewed in Paper 1 [[P1]]; what was missing prior to Paper 1 was any dynamical calculation within the framework.

Paper 1 [[P1]] supplied that missing step: a complete, benchmark-verified Wilson lattice QCD simulation was constructed within $\text{Cl}(6)$, demonstrating numerical equivalence with standard Wilson lattice QCD across gauge and fermionic observables. Section 5.6.4 of that paper gestured toward what is now the subject of the present work: *grade-decomposed observables*, defined in terms of the $\text{Cl}(6)$ grade structure and carrying $\text{SU}(3)$ representation-theoretic content that the standard four-component formulation cannot express.

This paper delivers what Section 5.6.4 of Paper 1 gestured at. We construct the grade decomposition of the pion correlator explicitly, prove a kinematic limit theorem that governs its free-field behavior, and implement it on Monte Carlo gauge ensembles. The motivating physical question is whether non-perturbative gauge dynamics—confinement, chiral symmetry breaking, the quantum vacuum—distribute spinor weight unequally among the four representation channels. The present work establishes the algebraic machinery and confirms the null result at heavy quark mass. The chiral-regime test of the central question is identified as future work.

1.2 Prior Work and the Gap

Three distinct bodies of literature are relevant to the present paper, and none overlaps with its essential contribution.

The first is the algebraic classification program in $\text{Cl}(6)$ and related division algebras [20–26]. Furey’s work [20–24] identifies the quantum numbers of a single Standard Model generation from the action of $\text{Cl}(6)$ on its minimal left ideal, and connects the $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ symmetry to the algebra’s automorphism structure. Stoica [26] extends related constructions. Dixon [25] works in the division algebra

context. None of these works computes dynamics, and none formulates a lattice theory.

The second is the Pati–Salam $SU(4)_{\text{color}}$ model [1] and the associated branching rules of the $SU(4) \supset SU(3) \times U(1)$ Levi decomposition, tabulated by Slansky [2]. These establish the representation-theoretic content of the (Γ_7, H) double grading at the group theory level; the present paper is the first work to implement this grading dynamically.

The third is the broader landscape of alternative representations for $SU(3)$ lattice gauge theory—loop-string-hadron formulations, tensor network encodings, qubit-compatible embeddings; see Paper 1 [[P1]] §1.2 for a full survey. None of these uses the Clifford algebra as the reformulation language, and none introduces grade-projected correlators.

The gap filled by this paper is the first dynamical implementation of the (Γ_7, H) grading: construction of the projectors P_r , proof of the kinematic limit theorem, and measurement of the grade fractions on thermalized gauge ensembles.

1.3 Scope and Organization

This paper makes three contributions. First, the algebraic construction of the four-channel grade decomposition of the pion correlator, including the Cartan hypercharge H as the grading operator and the complete set of projector identities (Secs. 2 and 4). Second, a kinematic limit theorem proving that free-field grade fractions equal $\dim(r)/8$ exactly, with a full algebraic proof from the $I_4 \otimes U(2)$ symmetry of the free Wilson–Dirac operator (Sec. 6.5). Third, the first numerical implementation on quenched $SU(3)$ gauge ensembles, with $N = 6$, $\beta \in \{4.5, 6.0\}$, and $m_0 = 0.05$ (Sec. 6.6).

This paper does not claim to have measured a dynamical grade-fractionation signal. The heavy-mass data is consistent with the kinematic theorem at the 10^{-4} level; the chiral-regime extension needed to test the central hypothesis is identified as Experiment B (Sec. 6.8).

The present work is deliberately restricted to the grade-decomposition observable. A broader programme—renormalized Gell–Mann–Oakes–Renner analysis and Banks–Casher spectral measurements—was considered but is not pursued here: the renormalized-GOR signal is dominated by additive Wilson mass renormalization and does not cleanly isolate the physics of interest, and the spectral-density measurement belongs to the chiral-regime extension (Experiment B, Sec. 6.8.2). The κ_c^{ren} determination of Sec. 5 is retained solely as a prerequisite for that extension, not as input to any renormalized-GOR fit.

Three limitations govern the current results: single lattice volume ($N = 6$), single lattice spacing ($\beta \in \{4.5, 6.0\}$ fixed), and quenched approximation throughout.

Conventions. Euclidean signature throughout. Spacetime indices $\mu, \nu \in \{1, 2, 3, 4\}$; $Cl(6)$ internal indices $i, j \in \{1, \dots, 6\}$. Wilson parameter $r = 1$ fixed; $\kappa = 1/[2(m_0 + 4r)]$. All conventions match Paper 1 [[P1]] exactly; that paper is the primary foundational reference for algebraic details not reprised here.

2 Algebraic Framework

2.1 Cl(6) Basis and Grade Structure

The Clifford algebra $\text{Cl}(6)$ is generated by six orthonormal basis vectors $\{e_1, \dots, e_6\}$ satisfying

$$e_i e_j + e_j e_i = 2\delta_{ij}, \quad i, j \in \{1, \dots, 6\}. \quad (1)$$

An 8×8 matrix representation is given by Pauli tensor products:

$$\begin{aligned} e_1 &= \sigma_x \otimes I_2 \otimes I_2, & e_2 &= \sigma_y \otimes I_2 \otimes I_2, \\ e_3 &= \sigma_z \otimes \sigma_x \otimes I_2, & e_4 &= \sigma_z \otimes \sigma_y \otimes I_2, \\ e_5 &= \sigma_z \otimes \sigma_z \otimes \sigma_x, & e_6 &= \sigma_z \otimes \sigma_z \otimes \sigma_y. \end{aligned} \quad (2)$$

All six are Hermitian and square to I_8 ; the anti-commutation relations (1) hold to maximum error 0.00×10^0 in IEEE 754 double precision. We refer to Paper 1 [[P1]] Appendix A for the complete 64-element basis enumeration and multiplication table. The grade decomposition

$$\text{Cl}(6) = \bigoplus_{k=0}^6 \text{Cl}^k(6), \quad \dim \text{Cl}^k(6) = \binom{6}{k}, \quad (3)$$

has total dimension 64. Of the seven grade subspaces, the relevant ones for this work are the bivectors $\text{Cl}^2(6)$ (dimension 15), which contain the Lie algebra of $\text{Spin}(6) \cong \text{SU}(4)$, and the pseudoscalar $\text{Cl}^6(6)$, spanned by $I_6 = e_1 e_2 e_3 e_4 e_5 e_6$ with $I_6^2 = -1$. The 8-dimensional minimal left ideal decomposes under $\text{SU}(3)$ as

$$\mathbf{8} = \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}_a \oplus \mathbf{1}_b, \quad (4)$$

where $\mathbf{3}$ and $\bar{\mathbf{3}}$ are the color triplet and anti-triplet and $\mathbf{1}_a, \mathbf{1}_b$ are two color singlets. The full quark space is $\mathbb{C}^{24} = \mathbb{C}^8 \otimes \mathbb{C}^3$, with $\mathbb{C}^8$ the $\text{Cl}(6)$ spinor space and $\mathbb{C}^3$ the standard QCD color space.

2.2 SU(3) Embedding via $\text{Spin}(6) \cong \text{SU}(4)$

The isomorphism chain

$$\text{SU}(3)_{\text{color}} \subset \text{SU}(4) \cong \text{Spin}(6) \subset \text{Cl}^+(6) \subset \text{Cl}(6) \quad (5)$$

embeds $\text{SU}(3)$ as the stabilizer subgroup of a preferred spinor $\xi_0 = (0, 0, 0, 1)^T \in \mathbb{C}^4$. The 15 bivectors $B_{ij} = (i/4)[e_i, e_j]$ span the Lie algebra of $\text{Spin}(6)$ and contain all eight $\text{SU}(3)$ generators as a subalgebra. Gell-Mann normalization $\text{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}$ is maintained; all 28 $\text{SU}(3)$ commutators are verified to maximum deviation 1.24×10^{-16} (see Paper 1 [[P1]] Appendix B). Independently, the bivector $\text{SU}(3)$ octet—derived as the stabilizer of the $H = -\frac{3}{2}$ eigenstate $|\xi_0\rangle$ within the 15-dimensional bivector space—is verified to close as a Lie algebra under the matrix commutator directly in the 8×8 representation (maximum closure residual 6.2×10^{-16} ; Appendix B). $\text{SU}(3)$ gauge links act as standard 3×3 unitary matrices on the $\mathbb{C}^3$ color factor and are unchanged by the $\text{Cl}(6)$ embedding. **SU(3) is embedded by construction, following the known isomorphism chain; it is not derived from $\text{Cl}(6)$.**

2.3 The Cartan Hypercharge H and Its Eigenspectrum

This section introduces algebraic content not present in Paper 1 [\[\[P1\]\]](#). The Cartan hypercharge of $\text{spin}(6)$ corresponding to the $\text{SU}(3) \times \text{U}(1)$ Levi decomposition is

$$H = B_{12} + B_{34} + B_{56}, \quad (6)$$

where $B_{ij} = (i/4)[e_i, e_j]$ are bivectors. In the tensor product representation [\(2\)](#), H is diagonal:

$$H = -\frac{1}{2}(\sigma_z \otimes I_2 \otimes I_2) - \frac{1}{2}(I_2 \otimes \sigma_z \otimes I_2) - \frac{1}{2}(I_2 \otimes I_2 \otimes \sigma_z) \quad (7)$$

$$= \frac{1}{2} \text{diag}(-3, -1, -1, +1, -1, +1, +1, +3). \quad (8)$$

The eigenspectrum is $\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\}$ with multiplicities $\{1, 3, 3, 1\}$, matching the $\text{SU}(3)$ representation dimensions in [\(4\)](#). Lagrange spectral projectors onto each H -eigenspace are

$$P_r = \prod_{h' \neq h_r} \frac{H - h' I_8}{h_r - h'}, \quad (9)$$

where h_r is the H -eigenvalue of representation r . The resulting projectors satisfy $\text{Tr}(P_3) = 3$, $\text{Tr}(P_{\bar{3}}) = 3$, $\text{Tr}(P_{1_a}) = 1$, $\text{Tr}(P_{1_b}) = 1$, and $\sum_r P_r = I_8$, all to machine precision. Idempotency $P_r^2 = P_r$ and orthogonality $P_r P_s = 0$ for $r \neq s$ also hold to machine precision. Full verification in [Appendix C](#).

2.4 The (Γ_7, H) Double Grading

The eight spinor basis states are simultaneously labeled by two commuting operators. The $\text{Spin}(6)$ chirality element

$$\Gamma_7 = iI_6 = i \cdot e_1 e_2 e_3 e_4 e_5 e_6 = \sigma_z \otimes \sigma_z \otimes \sigma_z \quad (10)$$

takes eigenvalues ± 1 . This is distinct from the 4D chiral operator $\gamma_5 = e_1 e_2 e_3 e_4 = -\sigma_z \otimes \sigma_z \otimes I_2$ used in the pion correlator. The commutativity $[\Gamma_7, H] = 0$ holds identically in the representation [\(2\)](#), and the four $\text{SU}(3)$ representations are uniquely labeled by the pair (Γ_7, H) :

Table 1: The (Γ_7, H) double grading of the spinor basis states. This is the standard $\text{SU}(4) \supset \text{SU}(3) \times \text{U}(1)$ Levi decomposition; see Ref. [\[2\]](#), Table 36.

Rep	Γ_7	H	dim	Basis states
1_a	+1	-3/2	1	$ \uparrow\uparrow\uparrow\rangle$
3	+1	+1/2	3	$ \uparrow\downarrow\downarrow\rangle, \downarrow\uparrow\downarrow\rangle, \downarrow\downarrow\uparrow\rangle$
$\bar{3}$	-1	-1/2	3	$ \uparrow\uparrow\downarrow\rangle, \uparrow\downarrow\uparrow\rangle, \downarrow\uparrow\uparrow\rangle$
1_b	-1	+3/2	1	$ \downarrow\downarrow\downarrow\rangle$

The (Γ_7, H) labeling is the standard $\text{SU}(4) \supset \text{SU}(3) \times \text{U}(1)$ Levi decomposition, tabulated by Slansky [\[2\]](#) (Table 36). With the normalization $Q_{B-L} = (2/3)H$,

the triplets carry baryon-minus-lepton charge $Q_{B-L} = \pm 1/3$ and the singlets carry $Q_{B-L} = \mp 1$, matching the Pati–Salam $SU(4)_{\text{color}}$ model [1]. The same decomposition arises in Furey’s algebraic classification program [20–24] as the identification of $SU(3)$ quantum numbers for a single generation of Standard Model fermions from the $Cl(6)$ minimal left ideal.

The key distinction for the dynamics of this paper is that $\Gamma_7 \neq \gamma_5$. The relationship $\Gamma_7 = \gamma_5 \cdot (ie_5e_6)$ shows that Γ_7 combines the 4D Lorentz chirality with the chirality of the internal (e_5, e_6) pair. Both Γ_7 and γ_5 commute with H and with all four projectors P_r , but the triplet subspace $\mathbf{3}$ contains states of both γ_5 eigenvalues—a fact that is fundamental to the analysis of Sec. 6.7.5.

3 Lattice Implementation

3.1 Gauge Ensembles and Parameters

All production calculations use hypercubic lattices with periodic boundary conditions in all four directions. The production volume for the grade decomposition calculations is $N = 6$ ($V = 1296$ sites). Two gauge couplings are employed: $\beta = 4.5$ (confined phase, $\langle P \rangle \approx 0.340$) and $\beta = 6.0$ (deconfined phase, $\langle P \rangle \approx 0.593$). The phase boundary at $\beta_c \approx 5.69$ was located in Paper 1 [[P1]] §4.1.2 and is not remeasured here.

Gauge configurations are generated by Metropolis updates at acceptance rate $\approx 50\%$. Thermalization was extended to $N_{\text{therm}} = 1000$ sweeps at $\beta = 4.5$ and $N_{\text{therm}} = 800$ sweeps at $\beta = 6.0$ (compared to $N_{\text{therm}} = 300$ sweeps in Paper 1 [[P1]]); this additional thermalization was adopted following the checkerboard correction described in Sec. 3.3. Decorrelation uses $N_{\text{decorr}} = 5$ sweeps. Production ensembles: $N_{\text{cfg}} = 20$ configurations per β value.

Table 2 reports the plaquette equilibration gate; both ensembles pass the $\pm 1\%$ gate at 1σ .

Table 2: Plaquette equilibration gate. Corrected even-odd sweep produces $\langle P \rangle$ within 1% of Paper 1 [[P1]] $N = 6$ values in both phases.

β	$\langle P \rangle$ measured	Paper 1 $N = 6$ $\langle P \rangle$	Deviation	Gate
4.5	0.34063 ± 0.00050	0.33816	+0.73%	✓ PASS
6.0	0.59580 ± 0.00073	0.59334	+0.41%	✓ PASS

3.2 Wilson–Dirac Operator

The Wilson–Dirac operator in $Cl(6)$ is unchanged from Paper 1 [[P1]] §3.3:

$$\begin{aligned}
 (D_W \Psi)(x) = (m_0 + 4r) \Psi(x) - \frac{1}{2} \sum_{\mu} \Big[(1 - re_{\mu}) U_{\mu}(x) \Psi(x + \hat{\mu}) \\
 + (1 + re_{\mu}) U_{\mu}^{\dagger}(x - \hat{\mu}) \Psi(x - \hat{\mu}) \Big],
 \end{aligned}
 \tag{11}$$

with $r = 1$ fixed and the action on $\mathbb{C}^8 \otimes \mathbb{C}^3$ per site given by $(1 \pm e_\mu) \otimes U_\mu(x)$. The bare quark mass parameter for the heavy-mass production runs is $m_0 = 0.05$, corresponding to $\kappa = 1/(2(m_0 + 4)) = 0.12346$.

The conjugate-gradient solver is applied to $D_W^\dagger D_W x = D_W^\dagger b$ with convergence criterion $\|r\|/\|b\| < 10^{-8}$ and maximum 500 iterations. Mean CG iterations were 18.95 at $\beta = 4.5$ and 21.85 at $\beta = 6.0$; there were zero CG failures in the $N_{\text{cfg}} = 20$ production ensembles at $m_0 = 0.05$. The pion propagator is built from a single point-to-all solve at the origin, spanning all 24 spin-color source components (8 spinor $\times$ 3 color); the solve is exact rather than stochastic, so each configuration yields one deterministic correlator. The three classes of measurement in this work accordingly use distinct source types: the Experiment A grade fractions use this exact point-to-all solve; the κ_c^{ren} /PCAC determination uses Z_2 stochastic sources; and the free-field and completeness diagnostics use a single Z_2 source, valid there because those identities are source-independent.

3.3 Even-Odd Checkerboard Sweep Correction

A methodological correction relative to some intermediate runs is documented here. Earlier runs employed a vectorized parallel Metropolis sweep that computed all staples simultaneously before proposing any updates. This approach violates detailed balance: once a proposal at site x is accepted, its neighbors' staples are modified, but those neighbors were already evaluated against pre-update values within the same sweep. The resulting bias scales with coupling strength and manifests as a systematic plaquette deficit—approximately 5.1% at $\beta = 4.5$ and 2.4% at $\beta = 6.0$, independent of sweep count.

The corrected implementation uses an even-odd checkerboard sweep (see Paper 1 [[P1]] §3.4 for the Metropolis update framework). For each direction μ and parity $p \in \{0, 1\}$: staples for all parity- p sites are computed using the current link configuration (including freshly-updated parity- $(1 - p)$ sites from the preceding half-step), and proposals and acceptances for parity- p sites are made in parallel. Within a parity class, no two sites share a same-direction, same-parity staple, so detailed balance holds exactly at each half-step. All production data in this paper use the corrected checkerboard sweep. We note that this correction retroactively suggests that some intermediate- β plaquette values reported in Paper 1 [[P1]] ($\beta = 5.0$ at 7% low, $\beta = 5.5$ at 5% low) may have the same origin and warrant re-examination.

4 Grade Projectors: Construction and Verification

4.1 Construction as Lagrange Spectral Projectors

The four grade projectors are constructed as Lagrange spectral projectors of the Cartan hypercharge H defined in (6). The four H -eigenvalues are $h_{\mathbf{1}_a} = -\frac{3}{2}$, $h_{\mathbf{3}} = +\frac{1}{2}$, $h_{\bar{\mathbf{3}}} = -\frac{1}{2}$, and $h_{\mathbf{1}_b} = +\frac{3}{2}$; the projector onto representation r is given by (9).

This construction guarantees idempotency $P_r^2 = P_r$ and orthogonality $P_r P_s = 0$

for $r \neq s$ by Lagrange interpolation theory, independently of the specific form of H . Completeness $\sum_r P_r = I_8$ follows from the Lagrange basis being a partition of unity over any finite set of distinct points. Full algebraic verification is reported in Appendix C, Table C.1.

The charge conjugation matrix C in $\text{Cl}(6)$ is constructed from first principles via the requirement $C e_i C^\dagger = -e_i^*$ for all $i \in \{1, \dots, 6\}$. It is not the standard 4×4 Dirac charge conjugation matrix; it must be derived from the 8×8 $\text{Cl}(6)$ basis (2). The resulting C satisfies $C P_3 C^\dagger = P_3$ and $C P_{1_a} C^\dagger = P_{1_b}$ exactly (residual 0.00×10^0); see Appendix B.

4.2 SU(3) Covariance

The projectors commute with all eight $\text{SU}(3)$ generators: $[T_a, P_r] = 0$ for all $a \in \{1, \dots, 8\}$ and all r . This is not a separately proved identity but follows automatically from the algebraic structure: H is a polynomial in the Cartan generators of $\text{Spin}(6)$, and the $\text{SU}(3)$ generators T_a lie in the subalgebra that commutes with H by definition of the Cartan decomposition. Numerically, $\text{SU}(3)$ covariance $U P_r U^\dagger = P_r$ is verified over 50 random $U \in \text{SU}(3)$ for all four projectors, with maximum deviation 2.44×10^{-15} (Table C.1, row C5).

4.3 Commutativity with γ_5

The commutativity $[\gamma_5, P_r] = 0$ for all r holds exactly (residual 0.00×10^0). This property is essential for the grade-projected pion correlator to be well-defined: without it, the operator $P_r \gamma_5 P_r$ would mix chiralities and $C_\pi^{(r)}(t)$ would not define a physical pseudoscalar channel. The commutativity follows from the observation that $\gamma_5 = e_1 e_2 e_3 e_4$ acts only on the first two tensor factors in (2), while H acts on all three; the commutator vanishes because γ_5 commutes with each B_{ij} appearing in H .

4.4 H -Grade Labeling via I_6

The pseudoscalar $I_6 = e_1 e_2 e_3 e_4 e_5 e_6$ acts on the two singlet subspaces as

$$\begin{aligned} I_6 P_{1_a} &= -i P_{1_a}, \\ I_6 P_{1_b} &= +i P_{1_b}, \end{aligned} \tag{12}$$

verified to residual 0.00×10^0 . This provides an alternative labeling of the singlet channels consistent with the (Γ_7, H) double grading of Sec. 2.4.

4.5 Complete Verification Table

The full projector algebraic verification is reported as Table C.1 (Appendix C). All five algebraic checks (C1–C5) and one stochastic check (S1) pass to machine precision. The $I_4 \otimes U(2)$ symmetry argument of Sec. 6.5 depends only on checks C1–C5; S1 is an additional stochastic verification of group-averaging behavior under the Haar measure.

5 Non-Perturbative κ_c^{ren} Determination

5.1 PCAC Method

The renormalized critical hopping parameter $\kappa_c^{\text{ren}}(\beta)$ is the value of κ at which the PCAC quark mass vanishes:

$$m_{\text{PCAC}}(\kappa) = \frac{C_{AP}(t)}{4 C_{PP}(t)}, \quad (13)$$

where $C_{AP}(t)$ and $C_{PP}(t)$ are the axial-pseudoscalar and pseudoscalar-pseudoscalar correlators, respectively, plateaued over $t \in [N/4, 3N/4]$. In the Cl(6) context, the axial current is $A_4^a(x, t) = \bar{\psi}(x)(e_4 \otimes I_3)(\gamma_5 \otimes I_3)(I_8 \otimes \lambda^a/2)\psi(x)$, which requires the tensor product structure of Paper 1 [[P1]] §2.3. The pseudoscalar density is $P^a(x, t) = \psi(x)(\gamma_5 \otimes I_3)(I_8 \otimes \lambda^a/2)\psi(x)$.

The zero crossing of $m_{\text{PCAC}}(\kappa)$ is located by Lagrange interpolation between the two κ values bracketing $m_{\text{PCAC}} = 0$, with the bracket selection requiring $\text{SNR} > 0.8$ on both endpoints to exclude noise-dominated crossings. Z_2 stochastic sources are used for the correlator estimates.

Method B (m_π^2 extrapolation via cosh fit) was attempted but proved unreliable on $N \leq 6$ lattices due to the insufficient plateau window at small volumes. All κ_c^{ren} values quoted in this section use Method A (PCAC) only.

Because Method B did not yield a usable determination, the κ_c^{ren} values reported here rest on the single PCAC method and carry no independent cross-check: the two-method consistency test that would normally validate a κ_c^{ren} determination could not be performed. Accordingly these values are used only to locate the chiral regime for Experiment B (Sec. 6.8.2); the m_π^2 -extrapolation cross-check is identified as follow-up work in Sec. 6.8.4.

5.2 Results

Block 1 of the computation produced partial results sufficient for the grade decomposition program. Two of six β values were determined; the remaining four are not required for Paper 2.

Monotonicity check: $\kappa_c^{\text{ren}}(\beta = 4.5) = 0.1468 < 0.1513 = \kappa_c^{\text{ren}}(\beta = 6.0)$, consistent with the physical expectation that weaker coupling (larger β) produces smaller additive mass renormalization and κ_c^{ren} closer to the free-field value 0.125. The large uncertainty on $\kappa_c^{\text{ren}}(\beta = 4.5)$ reflects genuine $O(a)$ discretization effects at strong coupling that are not yet resolved; this uncertainty will propagate as a systematic into any chiral-regime calculation using this value.

The present heavy-mass production runs at $m_0 = 0.05$ ($\kappa = 0.12346$) lie well below κ_c^{ren} at both β values and do not require the chiral-regime solver mitigations of Appendix D. The κ_c^{ren} values are provided as prerequisites for Experiment B (Sec. 6.8.2) and are referenced in Sec. 6.8.

Table 3: Non-perturbative κ_c^{ren} values used in this paper. The $\beta = 4.5$ value is systematic-dominated; see text.

β	κ_c^{ren}	Source and notes
4.5	0.1468 ± 0.0050	$N = 6$, Method A (PCAC); systematic-dominated. $N = 4/N = 6$ spread ($0.127 \rightarrow 0.147$) propagated as ± 0.005 systematic uncertainty.
6.0	0.1513 ± 0.0006	Weighted mean across four independent runs; 0.9σ from literature value 0.1518 ± 0.0005 (Paper 1 [[P1]] §5.5.1).

6 Grade Decomposition of the Pion Correlator

6.1 The Grade-Projected Pion Correlator

The standard pion correlator in the Cl(6) framework is

$$C_\pi(t) = \sum_{\mathbf{x}} \left\langle \text{Tr} \left[S(\mathbf{x}, t; \mathbf{0}, 0) \gamma_5 S^\dagger(\mathbf{x}, t; \mathbf{0}, 0) \gamma_5 \right] \right\rangle, \quad (14)$$

where $S(\mathbf{x}, t; \mathbf{0}, 0) = D_W^{-1}(\mathbf{x}, t; \mathbf{0}, 0)$ is the quark propagator (a 24×24 matrix per site pair in $\mathbb{C}^8 \otimes \mathbb{C}^3$), the trace is over both spinor and color indices, and γ_5 acts on the spinor sector only. The grade-projected pion correlator for representation r is defined by inserting the projector P_r into the spinor trace:

$$C_\pi^{(r)}(t) = \sum_{\mathbf{x}} \left\langle \text{Tr} \left[(P_r \otimes I_3) \gamma_5 (P_r \otimes I_3) \cdot S(\mathbf{x}, t; \mathbf{0}, 0) \gamma_5 S^\dagger(\mathbf{x}, t; \mathbf{0}, 0) \right] \right\rangle. \quad (15)$$

The commutativity $[\gamma_5, P_r] = 0$ of Sec. 4.3 ensures that P_r and γ_5 can be commuted freely, so $C_\pi^{(r)}(t)$ defines a valid pseudoscalar channel. The grade fraction is

$$f_r(t) \equiv \frac{C_\pi^{(r)}(t)}{C_\pi(t)}. \quad (16)$$

By the completeness identity $\sum_r P_r = I_8$, it follows algebraically that

$$\sum_r C_\pi^{(r)}(t) = C_\pi(t) \quad (\text{exact, algebraically}), \quad (17)$$

and therefore $\sum_r f_r(t) = 1$ on every configuration and at every timeslice. This identity provides a per-configuration sanity check: any violation indicates a bug in the projector construction or trace evaluation.

6.2 Algebraic Completeness Verification

The identity (17) is verified via the per-configuration relative residual

$$\delta(t) = \frac{|C_\pi(t) - \sum_r C_\pi^{(r)}(t)|}{|C_\pi(t)|}. \quad (18)$$

In the free-field case ($U_\mu = I_3$ everywhere, $N = 4$, $m_0 = 0.05$, single Z_2 source), $\delta_{\max} = 1.26 \times 10^{-16}$ over $t \in \{0, 1, 2, 3\}$ —six orders of magnitude below the target threshold of 10^{-10} . In the production ensembles ($N = 6$, $N_{\text{cfg}} = 20$, a single point-to-all solve per configuration spanning 24 spin-color components):

$$\begin{aligned} \delta_{\max}(\beta = 4.5) &= 5.371 \times 10^{-16} && (\text{max over 20 configs} \times 6 \text{ timeslices}), \\ \delta_{\max}(\beta = 6.0) &= 5.366 \times 10^{-16} && (\text{max over 20 configs} \times 6 \text{ timeslices}). \end{aligned}$$

Both values are consistent with floating-point roundoff at $O(\varepsilon_{\text{mach}} \times N_{\text{hops}})$. The identity (17) is verified to machine precision on every configuration in the production ensemble.

6.3 Charge Conjugation and γ_5 Identities

Three additional algebraic identities are verified to machine precision prior to the production runs (Appendix B):

- Charge conjugation pairing: $CP_3C^\dagger = P_3$ and $CP_{1_a}C^\dagger = P_{1_b}$, each to residual 0.00×10^0 .
- γ_5 commutativity: $[\gamma_5, P_r] = 0$ for all r , to residual 0.00×10^0 .
- I_6 action on singlets: $I_6 P_{1_a} = -i P_{1_a}$ and $I_6 P_{1_b} = +i P_{1_b}$, each to residual 0.00×10^0 .

These identities, together with the projector checks of Appendix C, establish the algebraic infrastructure to machine precision before any gauge configurations are introduced.

6.4 H Eigenvalue and Additional Verifications

The Cartan hypercharge H is verified to have eigenvalues $\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\}$ with multiplicities $\{1, 3, 3, 1\}$ in the explicit basis (2). The commutativity $[\Gamma_7, H] = 0$ holds identically. The commutativity of H with all eight $\text{SU}(3)$ generators,

$$[H, T_a] = 0 \quad \text{for } a \in \{1, \dots, 8\},$$

is verified numerically to maximum deviation $< 10^{-15}$. These verifications confirm that H generates the $\text{U}(1)$ factor of the $\text{SU}(3) \times \text{U}(1)$ Levi subgroup, as required by the representation-theoretic argument of Sec. 2.4.

6.5 The Kinematic Limit Theorem

The central theoretical result of this paper is the following theorem, which governs the free-field behavior of the grade fractions and provides the null hypothesis for Experiment B (Sec. 6.8.2).

Theorem 6.1 (Kinematic Limit). *Let $G_{\text{free}} = (D_{\text{W}}^{\text{free}})^{-1}$ be the free-field quark propagator (all gauge links $U_\mu = I_3$). Then*

$$f_r^{\text{free}}(t) = \frac{C_\pi^{(r)}{}_{\text{free}}(t)}{C_\pi^{\text{free}}(t)} = \frac{\dim(r)}{8} \quad (19)$$

exactly, for all $r \in \{3, \bar{3}, 1_a, 1_b\}$ and all timeslices t . In the interacting theory, the same identity holds in the ensemble limit

$$\lim_{N_{\text{cfg}} \rightarrow \infty} \langle f_r(t) \rangle = \frac{\dim(r)}{8}, \quad (20)$$

enforced by SU(3) color symmetry and the hypercubic symmetry of the ensemble measure.

Proof. The proof requires two ingredients. The first is the commutativity $[\gamma_5, P_r] = 0$ established in Sec. 4.3: without it, the grade-projected pion operator $P_r \gamma_5 P_r$ would not define a valid pseudoscalar channel. The second concerns the Hermitian matrix

$$\tilde{Q}(t) = \sum_{\mathbf{x}} (\gamma_5 \otimes I_3) G(\mathbf{x}, t; \mathbf{0}, 0) (\gamma_5 \otimes I_3) G^\dagger(\mathbf{x}, t; \mathbf{0}, 0), \quad (21)$$

whose grade-projected traces give the channel correlators, $C_\pi^{(r)} = \text{Tr}[(P_r \otimes I_3) \tilde{Q}]$ and $C_\pi = \text{Tr}[\tilde{Q}]$. Since each P_r is diagonal in the (Γ_7, H) grade basis (in which H is diagonal), these traces depend only on the diagonal of $\tilde{Q}$ in that basis. We show that for the free field this diagonal is constant:

$$[\tilde{Q}_{\text{free}}(t)]_{\text{grade diag}} = c(t) I_{24}. \quad (22)$$

Given (22),

$$C_\pi^{(r)} = c \text{Tr}[P_r \otimes I_3] = 3c \dim(r), \quad C_\pi = 24c,$$

and $f_r = \dim(r)/8$ follows immediately, for all r and all t .

It remains to establish (22). In the tensor-product basis of Sec. 2.1 the spacetime generators satisfy $e_\mu = M_\mu \otimes I_2$ for all $\mu \in \{1, 2, 3, 4\}$: the third Clifford qubit factor is always I_2 . Hence $D_{\text{W}}^{\text{free}}$ commutes with every $I_4 \otimes U'' \otimes I_3$ for $U'' \in \text{U}(2)$, and by Schur's lemma ($\text{U}(2)$ acts irreducibly on the third $\mathbb{C}^2$) any matrix commuting with all such elements has the form $X \otimes I_2$. In the free theory the propagator is color-diagonal ($U_\mu = I_3$), so $X = A_4 \otimes I_3$, with A_4 acting on the four-dimensional (q_1, q_2) subspace. Summing over spatial sinks at fixed t projects onto zero spatial momentum, where the residual invariance is the *spatial-cubic group* $\times$ *time reflection*, not the full hypercubic

group. Spatial-cubic invariance removes the spatial generators e_2, e_3, e_4 (which form a cubic vector and average to zero) together with all tensor structures, leaving

$$A_4(t) = \alpha(t) I_4 + \beta(t) e_1, \quad (23)$$

where $e_1 = \sigma_x \otimes I_2$ is the time-direction generator. The term βe_1 need not vanish—time reflection forces $\beta(t)$ to be odd in t , so it vanishes only at the fixed timeslices $t = 0$ and $t = L/2$ —but e_1 has *identically zero diagonal* in the grade basis and so contributes nothing to the diagonal of $\tilde{Q} = A_4 \otimes I_2 \otimes I_3$, which is therefore $\alpha(t) I_{24}$. This establishes (22); the grade-fraction result rests only on this zero-diagonal property and holds whether or not $\beta = 0$.

The interacting identity (20) is a distinct, ensemble-averaged statement: it follows from applying the same SU(3) color and spatial-cubic symmetries to $\langle \tilde{Q} \rangle$, and holds for $\langle f_r \rangle$ in the $N_{\text{cfg}} \rightarrow \infty$ limit rather than configuration-by-configuration. $\square$

The physical content of the theorem is that the free Wilson–Dirac operator generates no dynamics in the e_5, e_6 directions: those generators appear nowhere in D_W^{free} . The grade decomposition is defined by $H = B_{12} + B_{34} + B_{56}$, and the bivector $B_{56} = \frac{i}{4}[e_5, e_6]$ is algebraically orthogonal to everything D_W^{free} can do.

Numerical verification. The theorem is verified at machine precision on the free lattice ($U_\mu = I_3$ everywhere, $N = 4$, $m_0 = 0.05$, single Z_2 source):

Grade	f_r measured	$\dim(r)/8$ expected	Residual
3	0.375000	0.375000	8.5×10^{-15}
$\bar{3}$	0.375000	0.375000	8.5×10^{-15}
1_a	0.125000	0.125000	8.4×10^{-15}
1_b	0.125000	0.125000	8.4×10^{-15}

Maximum residual from the expected free-field values: $O(\varepsilon_{\text{mach}})$ at all timeslices.

6.6 Experiment A: Heavy-Mass Production Results

Experiment A measures the grade fractions on thermalized quenched production ensembles at $N = 6$, $m_0 = 0.05$, $\beta \in \{4.5, 6.0\}$, with $N_{\text{cfg}} = 20$ configurations per phase. Parameters are as in Sec. 3.1; the bare quark mass $m_0 = 0.05$ corresponds to $\kappa = 0.123457$, placing the quarks well below κ_c^{ren} at both β values and deep in the heavy-mass regime ($m_{\text{phys}} \approx 0.76$ in lattice units).

Maximum deviation from the free-field reference at $t = 1$: $\approx 8.2 \times 10^{-5}$ at $\beta = 4.5$. All four phase differences are consistent with zero at 0.6σ significance.

The charge-conjugation pairing is manifest in the data: $\Delta f_3 = -\Delta f_{\bar{3}}$ and $\Delta f_{1_a} = -\Delta f_{1_b}$ to within statistical precision, as required by the identity $CP_3 C^\dagger = P_{\bar{3}}$ established in Sec. 6.3. The four σ -values group into two exactly-equal pairs: $\sigma(f_{1_a}) = \sigma(f_3)$ and $\sigma(f_{1_b}) = \sigma(f_{\bar{3}})$ hold to machine precision, forced configuration-by-configuration by the exact identity of Proposition 6.2 ($f_{1_a} = \frac{1}{2} - f_3$, $f_{1_b} = \frac{1}{2} - f_{\bar{3}}$); the remaining equality $\sigma(f_3) \approx \sigma(f_{\bar{3}})$ holds only statistically, by the ensemble-level charge-conjugation

Table 4: Grade fractions f_r at $t = 1$. Free-field reference: $f_r = \dim(r)/8 \in \{0.375, 0.375, 0.125, 0.125\}$. $\Delta \equiv f_r(\beta=4.5) - f_r(\beta=6.0)$. Errors are statistical, from a jackknife over the $N_{\text{cfg}} = 20$ gauge configurations (one point-to-all solve per configuration).

Grade	Rep	$f_r(\beta = 4.5)$	$f_r(\beta = 6.0)$	Δ	σ
3	Triplet	0.374918 ± 0.000079	0.374968 ± 0.000028	-0.000049 ± 0.000084	0.6
$\bar{3}$	Anti-triplet	0.375081 ± 0.000080	0.375032 ± 0.000028	$+0.000049 \pm 0.000085$	0.6
1_a	Singlet ($-i$)	0.125082 ± 0.000079	0.125032 ± 0.000028	$+0.000049 \pm 0.000084$	0.6
1_b	Singlet ($+i$)	0.124919 ± 0.000080	0.124968 ± 0.000028	-0.000049 ± 0.000085	0.6
Sum	Total	1.000000	1.000000	0.000000	—

relation $\langle f_3 \rangle = \langle f_{\bar{3}} \rangle$. Per configuration the four fractions therefore carry exactly two independent degrees of freedom, reducing to the single C-even observable f_{triplet} only in the ensemble average (Sec. 6.7.5).

t-dependence. The grade fractions at all timeslices $t \in \{0, \dots, 5\}$ are provided in the extended Table 6.1 of the Data Tables. Errors increase rapidly with t because the pion correlator decays as $e^{-m_\pi t}$ with $m_\pi \approx 3.4$ in lattice units: by $t = 3$, the signal has fallen by $e^{-10} \approx 5 \times 10^{-5}$. An apparent 3.6σ deviation in f_3 at $\beta = 6.0$, $t = 3$ is noise-dominated with $N_{\text{cfg}} = 20$ and does not constitute a physical signal; this is documented further in Sec. 6.7.5. At $t = 0$, the grade fractions satisfy $f_r \approx \dim(r)/8$ to within $\approx 3 \times 10^{-6}$, consistent with the short-range nature of the propagator at small t .

6.7 Discussion

6.7.1 The kinematic limit theorem

The result $f_r \approx \dim(r)/8$ observed throughout Table 4 is not a counting argument. It is a theorem with specific algebraic content (Theorem 6.1, Sec. 6.5), and identifying that content determines what the small deviations measure.

The physical content is that $D_{\text{W}}^{\text{free}}$ generates no dynamics in the e_5, e_6 directions: those generators appear nowhere in $D_{\text{W}}^{\text{free}}$. The grade decomposition is defined by $H = B_{12} + B_{34} + B_{56}$, and the bivector $B_{56} = \frac{i}{4}[e_5, e_6]$ is algebraically orthogonal to everything $D_{\text{W}}^{\text{free}}$ can do. Uniform distribution across H -eigenspaces follows not from any symmetry between e_5, e_6 and the spacetime generators, but from the absence of any coupling between $D_{\text{W}}^{\text{free}}$ and the internal directions that define the grading.

6.7.2 The (Γ_7, H) double grading and its place in the literature

Each grade projector is labeled by a pair of eigenvalues of two commuting operators. The Spin(6) chirality $\Gamma_7 = iI_6$, where $I_6 = e_1 e_2 e_3 e_4 e_5 e_6$ is the Cl(6) pseudoscalar, takes eigenvalues ± 1 and in the tensor product representation reads $\Gamma_7 = \sigma_z \otimes \sigma_z \otimes \sigma_z$. The

Cartan hypercharge $H = B_{12} + B_{34} + B_{56}$ takes eigenvalues $\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\}$ with multiplicities $\{1, 3, 3, 1\}$. Both operators are diagonal in the standard computational basis of $\mathbb{C}^8$, and their commutativity $[\Gamma_7, H] = 0$ follows immediately. The four representations are uniquely identified by the pair (Γ_7, H) :

Rep	Γ_7	H	dim	Basis states
$\mathbf{1}_a$	+1	$-3/2$	1	$ \uparrow\uparrow\uparrow\rangle$
$\mathbf{3}$	+1	$+1/2$	3	$ \uparrow\downarrow\downarrow\rangle, \downarrow\uparrow\downarrow\rangle, \downarrow\downarrow\uparrow\rangle$
$\bar{\mathbf{3}}$	-1	$-1/2$	3	$ \uparrow\uparrow\downarrow\rangle, \uparrow\downarrow\uparrow\rangle, \downarrow\uparrow\uparrow\rangle$
$\mathbf{1}_b$	-1	$+3/2$	1	$ \downarrow\downarrow\downarrow\rangle$

A distinction central to the interpretation is that $\Gamma_7 \neq \gamma_5$. The 4D chiral operator used in the pion correlator is $\gamma_5 = e_1 e_2 e_3 e_4 = -\sigma_z \otimes \sigma_z \otimes I_2$, while the Spin(6) chirality is $\Gamma_7 = \sigma_z \otimes \sigma_z \otimes \sigma_z$. Their relationship is $\Gamma_7 = \gamma_5 \cdot (ie_5 e_6)$, reflecting the fact that Γ_7 combines the 4D Lorentz chirality with the chirality of the internal (e_5, e_6) pair. Both commute with H and with all four projectors P_r , but their action on the spinor space differs: the triplet and antitriplet subspaces are each invariant under γ_5 but are not γ_5 eigenspaces—a point returned to in Sec. 6.7.5.

The (Γ_7, H) double grading is a known structure in the representation theory of SU(4). The operator H generates the U(1) factor of the Levi subgroup $\text{SU}(3) \times \text{U}(1) \subset \text{SU}(4)$, where $\text{SU}(3) = \text{Stab}_{\text{SU}(4)}(\xi_0)$ is the stabilizer of the preferred spinor direction $\xi_0 = (0, 0, 0, 1)^T$ that defines the color embedding. With the normalization $Q_{B-L} = (2/3)H$, the triplet/antitriplet carry $Q_{B-L} = \pm 1/3$ and the singlets carry $Q_{B-L} = \mp 1$, matching the baryon-minus-lepton charges of the Pati–Salam $\text{SU}(4)_{\text{color}}$ model [1]. The complete $\text{SU}(4) \supset \text{SU}(3) \times \text{U}(1)$ branching table— $4 \rightarrow 3_{+1/3} \oplus 1_{-1}$ and $\bar{4} \rightarrow \bar{3}_{-1/3} \oplus 1_{+1}$ —is tabulated in Ref. [2] (Table 36). In Furey’s algebraic classification program [20–24], the same decomposition identifies the SU(3) quantum numbers of a single generation of Standard Model fermions from the Cl(6) minimal left ideal.

The contribution of the present work is the first dynamical implementation of this grading. The projectors P_r are constructed as Lagrange spectral projectors of H , verified to satisfy completeness, idempotency, orthogonality, SU(3) covariance, and γ_5 -commutativity to residual $\leq 2.4 \times 10^{-15}$ (Table C.1), and used to decompose a physical pion correlator on a Monte Carlo gauge ensemble. The algebraic completeness identity $\sum_r C_\pi^{(r)}(t) = C_\pi(t)$ holds to $\delta_{\text{max}} = 5.4 \times 10^{-16}$ on every configuration. We emphasize that the structural analogy between the singlet subspaces $\mathbf{1}_a, \mathbf{1}_b$ and lepton quantum numbers in the Furey program does not apply in the dynamical QCD context of this work: here, color is carried by the separate $\mathbb{C}^3$ factor of the full quark space $\mathbb{C}^{24} = \mathbb{C}^8 \otimes \mathbb{C}^3$, and the grade projectors decompose the spinor sector $\mathbb{C}^8$ only.

6.7.3 Interpretation of the Experiment A deviations

The production data of Table 4 show grade fractions consistent with $\dim(r)/8$ at the 10^{-4} level. All phase differences $\Delta f_r = f_r(\beta=4.5) - f_r(\beta=6.0)$ are consistent with zero at 0.6σ significance. We identify three candidate mechanisms for the observed

deviations from free-field values and assess their relative importance at the current parameter point.

The primary driver is gauge-link color-spin entanglement. The Wilson hopping term $(1 \pm e_\mu) \otimes U_\mu(x)$ does not preserve H -eigenspaces: one finds $[H, e_\mu] \neq 0$ for all $\mu \in \{1, 2, 3, 4\}$. Explicitly,

$$[H, e_1] = \left[-\frac{1}{2}\sigma_z \otimes I_2 \otimes I_2, \sigma_x \otimes I_2 \otimes I_2 \right] = -i\sigma_y \otimes I_2 \otimes I_2 \neq 0, \quad (24)$$

and analogously for $\mu = 2, 3, 4$. Since P_r is a polynomial in H , it follows that $[P_r, D_W] \neq 0$: every lattice hop mixes grade channels, and the propagator $G = D_W^{-1}$ accumulates this mixing with each step. On a single gauge configuration, $\tilde{Q}_{\text{int}}$ is a full 24×24 matrix rather than a multiple of I_{24} , and the grade fractions can deviate from $\dim(r)/8$. In the ensemble limit, $\text{SU}(3)$ and hypercubic symmetry restore $\langle \tilde{Q}_{\text{int}} \rangle \propto I_{24}$, but this convergence is slow: with $N_{\text{cfg}} = 20$, the residual per-configuration entanglement is incompletely averaged.

Subleading contributions arise from the Wilson term's explicit $O(a)$ breaking of Lorentz symmetry, which introduces a small non-isotropic correction to $\tilde{Q}_{\text{int}}$ that is independent of N_{cfg} and vanishes only in the continuum limit. Genuine non-perturbative vacuum dressing of grade channels—mechanism (c), the target of Experiment B—is suppressed at $m_0 = 0.05$, where $m_{\text{phys}} \approx 0.76$ in lattice units and chiral effects scale as $(m_\pi/\Lambda_{\text{QCD}})^2$.

The observed sign pattern is consistent with the primary mechanism. At $t = 1$ and $\beta = 4.5$, the triplet fraction is depleted ($\Delta f_3 \approx -8.2 \times 10^{-5}$) while the singlet- a fraction is correspondingly enhanced ($\Delta f_{1_a} \approx +8.2 \times 10^{-5}$). This direction is expected from a perturbative $O(g^2)$ analysis: gauge-link dressing couples preferentially to color-charged spinor channels (the triplets), transferring a small fraction of weight toward the color-neutral singlet subspaces.

The t -dependence carries additional diagnostic information but is currently noise-limited. Deviations grow from $\sim 10^{-4}$ at $t = 1$ to $\sim 7 \times 10^{-4}$ at $t = 3$ at $\beta = 6.0$, reflecting the accumulation of grade mixing over additional hops as expected from (24). However, at $m_0 = 0.05$ the pion correlator is suppressed as $e^{-m_\pi t}$ with $m_\pi \approx 3.4$ in lattice units, so $C_\pi(t=3) \sim e^{-10} C_\pi(0)$. The apparent 3.6σ deviation at $t = 3$ is signal-to-noise dominated with $N_{\text{cfg}} = 20$ and does not constitute a physical signal at current statistics; resolving the t -profile reliably requires $N_{\text{cfg}} \gtrsim 200$.

6.7.4 Channel-pair structure and the C-symmetry reduction

The algebraic charge-conjugation pairing $CP_3 C^\dagger = P_3$ and $CP_{1_a} C^\dagger = P_{1_b}$ (Appendix B), together with the completeness constraint $\sum_r f_r = 1$, determines the observable structure precisely.

In the ensemble limit with the C-symmetric Wilson gauge action and a C-symmetric point source, charge conjugation enforces $\langle f_3 \rangle = \langle f_{\bar{3}} \rangle$ and $\langle f_{1_a} \rangle = \langle f_{1_b} \rangle$. Combined with completeness, this reduces four apparent observables to one independent C-even quantity:

$$f_{\text{triplet}} \equiv f_3 + f_{\bar{3}}, \quad (25)$$

with $f_{\text{singlet}} = 1 - f_{\text{triplet}}$ determined by completeness. The two C-odd combinations

$$A_{\text{color}} = f_3 - f_{\bar{3}}, \quad A_{\text{singlet}} = f_{1_a} - f_{1_b}, \quad (26)$$

both vanish in the $N_{\text{cfg}} \rightarrow \infty$ ensemble limit for $\theta = 0$. Their observed magnitudes ($\sim 1.6 \times 10^{-4}$ at $N_{\text{cfg}} = 20$) are finite-sample fluctuations expected to scale as $N_{\text{cfg}}^{-1/2}$, and they serve as self-consistency checks rather than physics signals. We note explicitly that A_{color} is C-odd and parity-even: it is not a probe of parity violation. A nonzero ensemble-average A_{color} would require C violation; the θ -term, being C-even, cannot generate it, and the present calculations are C-symmetric. By Proposition 6.2 this is moot in any case, since $A_{\text{color}} = -A_{\text{singlet}}$ exactly per configuration.

Proposition 6.2 (Spin(6) chirality is dynamically frozen). *On every gauge configuration and at every timeslice,*

$$C_{\pi}^{(\Gamma_7=+1)}(t) = C_{\pi}^{(\Gamma_7=-1)}(t), \quad \text{hence} \quad f_{(\Gamma_7=+1)}(t) = f_{(\Gamma_7=-1)}(t) = \frac{1}{2}, \quad (27)$$

so that $A_{\Gamma_7} \equiv A_{\text{color}} + A_{\text{singlet}} = 0$ identically. The result is independent of β , of t , and of the gauge field.

Proof. The Wilson–Dirac operator is built only from the spacetime generators $e_1, \dots, e_4$, each acting as $M_{\mu} \otimes I_2$ on the third Clifford qubit (Sec. 2.1), while the gauge links multiply the color index alone. The third qubit is therefore inert and the propagator factorizes as $G = \hat{G} \otimes I_2$ on that qubit on *every* configuration, free or interacting—a property unquenching cannot remove, as it alters the gauge measure but not this operator content. The γ_5 -dressed correlator bilinear $M(\mathbf{x}, t) = (\gamma_5 \otimes I_3) G (\gamma_5 \otimes I_3) G^{\dagger}$ (the per-configuration summand of $\hat{Q}$) inherits the same factorization, $M = \hat{M} \otimes I_2$, since $\gamma_5 = \hat{\gamma}_5 \otimes I_2$ likewise leaves the third qubit untouched. With $\Gamma_7 = \sigma_z \otimes \sigma_z \otimes \sigma_z$, the grade-summed chiral asymmetry factorizes across the third qubit,

$$C_{\pi}^{(\Gamma_7=+1)} - C_{\pi}^{(\Gamma_7=-1)} = \sum_{\mathbf{x}} \text{Tr}[(\Gamma_7 \otimes I_3) M] = \sum_{\mathbf{x}} \text{Tr}_{12c}[(\sigma_z \otimes \sigma_z \otimes I_3) \hat{M}] \text{Tr}_{q_3}[\sigma_z],$$

and the factor $\text{Tr}_{q_3}[\sigma_z] = 0$ annihilates it on every configuration. Dividing by C_{π} and using the per-configuration completeness $\sum_r f_r = 1$ gives $f_{(\Gamma_7=\pm 1)} = \frac{1}{2}$. $\square$

This identity is structurally stronger than, and must not be conflated with, the charge-conjugation relation. It fixes the *sum* $A_{\text{color}} + A_{\text{singlet}}$ to zero on *every* configuration: the measured residual is $\max |A_{\Gamma_7}| \leq 3.9 \times 10^{-16}$ across both ensembles, all configurations and timeslices, i.e. $A_{\text{color}} = -A_{\text{singlet}}$ *exactly*, not “to within statistical precision.” By contrast the individual asymmetries are each nonzero on a given configuration (per-configuration $|f_3 - f_{\bar{3}}|$ has rms $\approx 1.1 \times 10^{-3}$); only their *ensemble averages* vanish, by the genuinely statistical charge-conjugation relation $\langle f_3 \rangle = \langle f_{\bar{3}} \rangle$, $\langle f_{1_a} \rangle = \langle f_{1_b} \rangle$. The Spin(6)-chirality direction therefore carries no dynamical information whatsoever (Sec. 6.7.5).

The surviving observable $f_{\text{triplet}} = f_3 + f_{\bar{3}}$ measures the fraction of pion correlator weight in the color-carrying spinor channels. Its free-field value is 3/4, verified in both phases at the 0.6σ level (Table 4). The phase difference $\delta f_{\text{triplet}} \equiv f_{\text{triplet}}(\beta=4.5) -$

$f_{\text{triplet}}(\beta=6.0)$ is the target observable of Experiment B: it measures whether the gauge vacuum distributes spinor weight differently between color-carrying and color-neutral internal channels at the chiral point, and constitutes the grade-decomposed analog of ΔA from Paper 1 [[P1]].

6.7.5 The grade decomposition as a Cl(6)-specific structure

A natural question is whether the grade-projected correlator $C_\pi^{(r)}(t)$ could be computed in the standard four-component Wilson lattice QCD formulation by inserting an appropriate projector. The answer is no, for a specific algebraic reason.

The grade projectors are Lagrange spectral projectors of $H = B_{12} + B_{34} + B_{56}$, where

$$B_{56} = \frac{i}{4}[e_5, e_6]. \quad (28)$$

The bivector B_{56} requires the internal Clifford generators e_5 and e_6 , which belong to Cl(6) but have no counterpart in the standard four-component Dirac algebra Cl(4). The generators available in the standard formulation are $\{\gamma_1, \gamma_2, \gamma_3, \gamma_4\}$ and their products; without e_5 and e_6 , the bivector B_{56} cannot be formed, H cannot be constructed, and therefore P_r cannot be defined.

A further obstruction arises from the structure of the γ_5 chirality decomposition. The triplet subspace **3** (the $H = +1/2$ eigenspace) contains two basis states with $\gamma_5 = +1$ and one with $\gamma_5 = -1$; the antitriplet $\bar{\mathbf{3}}$ similarly has mixed γ_5 eigenvalues. The explicit assignments are:

State	Rep	γ_5	Γ_7	H
$ \uparrow\downarrow\downarrow\rangle$	3	+1	+1	+1/2
$ \downarrow\uparrow\downarrow\rangle$	3	+1	+1	+1/2
$ \downarrow\downarrow\uparrow\rangle$	3	-1	+1	+1/2
$ \uparrow\uparrow\downarrow\rangle$	$\bar{\mathbf{3}}$	-1	-1	-1/2
$ \uparrow\downarrow\uparrow\rangle$	$\bar{\mathbf{3}}$	+1	-1	-1/2
$ \downarrow\uparrow\uparrow\rangle$	$\bar{\mathbf{3}}$	+1	-1	-1/2
$ \uparrow\uparrow\uparrow\rangle$	1_a	-1	+1	-3/2
$ \downarrow\downarrow\downarrow\rangle$	1_b	-1	-1	+3/2

No combination of standard Dirac projectors—chiral projectors $(1 \pm \gamma_5)/2$, helicity projectors, or spin projectors $\sigma_{\mu\nu}$ —can isolate a grade subspace, because the grade structure cuts across the γ_5 sectors. This is not an artifact of the chosen basis; it is a consequence of the tensor product structure of the eight-dimensional spinor and cannot be removed by any change of basis within Cl(4).

The formal possibility of reconstructing $C_\pi^{(r)}(t)$ in the standard formulation—by reading off the explicit 8×8 matrix for P_r from the Cl(6) calculation and using it as an ad hoc operator—does not constitute an independent standard QCD construction. The matrix would be algebraically unmotivated without H ; its gauge invariance $[T_a, P_r] = 0$ would require separate verification; and its physical interpretation as a projector onto a specific SU(3) representation subspace of the spinor would be inaccessible. Moreover,

the Cl(6) quark space $\mathbb{C}^{24} = \mathbb{C}^8 \otimes \mathbb{C}^3$ does not embed naturally into the standard quark space $\mathbb{C}^{12} = \mathbb{C}^4 \otimes \mathbb{C}^3$: the factor-of-two mismatch in spinor dimension means P_r cannot be directly expressed as a matrix in the 12×12 space without first specifying an embedding of $\mathbb{C}^4$ into $\mathbb{C}^8$, a choice that is itself non-canonical and requires the Cl(6) structure to motivate.

The Cl(6) formulation provides three things that the standard formulation cannot: the internal generators e_5, e_6 that make H and P_r definable; the algebraic guarantees— $[T_a, P_r] = 0$ (gauge invariance), $[\gamma_5, P_r] = 0$ (chiral compatibility), $\sum_r P_r = I_8$ (completeness)—all following automatically from the algebra without separate proof; and the (Γ_7, H) interpretation that identifies the SU(3) representation content of each channel. The grade decomposition of the pion correlator is accordingly not a renaming of a standard QCD observable, but a structure that is natural and algebraically guaranteed in Cl(6) and has no natural standard QCD expression.

It is worth distinguishing two senses in which this structure might be called novel. The grade decomposition is *definable* only in Cl(6): the construction above requires the internal generators e_5, e_6 and is independent of any dynamics, so in that structural sense it is not a relabeling of a standard color projection. Whether it additionally *carries* physical information inaccessible to standard QCD is a distinct question, and one the present heavy-mass data do not settle: at $m_0 = 0.05$ the measured fractions remain pinned to their kinematic values $\dim(r)/8$ (Sec. 6.6), so at this parameter point the decomposition exhibits no resolvable dynamical content. Testing the informational sense—whether the grade fractions resolve gauge dynamics not already captured by ordinary color and chiral observables—is the purpose of Experiment B (Sec. 6.8.2).

Proposition 6.2 sharpens this question and bounds it from one side. The Spin(6)-chirality direction—the (e_5, e_6) content carried by Γ_7 —is *provably* information-free: $f_{(\Gamma_7=+1)} = f_{(\Gamma_7=-1)} = \frac{1}{2}$ on every configuration, independent of β , of t , and of the gauge field, precisely because e_5, e_6 never enter D_W . Whatever dynamical content the grade fractions can ever resolve must therefore reside entirely in the splitting *within* a fixed Γ_7 sector—the triplet-versus-singlet H -splitting of the (Γ_7, H) double grading, a γ_5 -type four-dimensional structure dressed by color. This is exactly the single C-even observable $f_{\text{triplet}} = f_3 + f_{\bar{3}}$ isolated in Sec. 6.7.4, and it is the target of Experiment B. The exact identity thus does not weaken the novelty claim: it removes a direction in which no new information could ever appear, and concentrates the informational question onto the one observable capable of carrying it.

6.8 Outlook

6.8.1 The random-walk hypothesis for the chiral regime

The algebraic result (24), $[H, e_\mu] \neq 0$, implies that every spacetime hop of the quark shifts its H -eigenvalue. More precisely, each application of e_μ for $\mu \in \{1, 2, 3, 4\}$ acts as a ± 1 step on the H -eigenvalue ladder $\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\}$, and the accumulated product $G = D_W^{-1}$ therefore constitutes a random walk in the grade space of the eight-dimensional spinor. The stationary distribution of this walk—the long-time limit of the grade fraction $f_r(t)$ —is determined by the gauge dynamics, not by free-field

counting.

At heavy quark mass ($m_0 = 0.05$, $m_{\text{phys}} \approx 0.76$), the propagator is short-range and the walk effectively terminates after $O(1)$ hops. The gauge-link weighting of each step is small ($|U_\mu - I_3|$ is perturbative in g^2), the mixing is nearly symmetric, and the free-field theorem controls the grade fractions: $f_r \approx \dim(r)/8$ at the 10^{-4} level, as confirmed by Experiment A.

At $\kappa \rightarrow \kappa_c^{\text{ren}}$, the pion mass $m_\pi \rightarrow 0$ and the propagator becomes long-range. For $t m_\pi \ll 1$ the kinematic regime persists and the free-field theorem still governs the grade fractions. For $t m_\pi \sim 1$, however, the propagator has traversed $O(m_\pi^{-1})$ lattice spacings and accumulated $O(m_\pi^{-1})$ gauge-link-weighted H -mixing steps. The grade fractions $f_r(t)$ will then curve from their free-field values toward an asymptotic limit determined by the ground-state pion wavefunction in the grade space: $f_r(t \rightarrow \infty) \rightarrow Z_\pi^{(r)}/Z_\pi$.

The central hypothesis motivating Experiment B is that the confined gauge vacuum ($\beta = 4.5$) biases this walk toward the color-triplet H -eigenspaces more strongly than the deconfined vacuum ($\beta = 6.0$), producing a nonzero phase difference

$$\delta f_{\text{triplet}} \equiv f_{\text{triplet}}(\beta = 4.5) - f_{\text{triplet}}(\beta = 6.0) \neq 0 \quad (29)$$

near κ_c^{ren} . The motivating argument has two parts, only the first of which is established.

The first part is a standard lattice-QCD expectation, not a measurement made here. In the chiral regime that Experiment B would probe, the strongly disordered gauge vacuum of the confined phase is expected to support a denser spectrum of near-zero Dirac modes than the deconfined phase: by analogy with the Banks–Casher mechanism $\Sigma = \pi\rho(0)$ [3], in which the chiral condensate Σ is proportional to the spectral density at the origin, $|\Sigma|$ would be larger at $\beta = 4.5$ than at $\beta = 6.0$. Such low modes would dominate the long-range pion propagator near κ_c^{ren} through the spectral decomposition $G(x, t; 0, 0) = \sum_n \psi_n(x, t) \psi_n^\dagger(0, 0)/\lambda_n$, in which the small- λ_n contributions are weighted most heavily. This chain is conventional lattice QCD; the present heavy-mass work neither reaches that regime nor measures $\rho(0)$.

The second part is conjectural and is precisely what Experiment B is designed to test: that the low Dirac modes, when projected onto the Cl(6) spinor space, carry preferential weight in the color-triplet H -eigenspaces ($H = \pm 1/2$) relative to the singlet H -eigenspaces ($H = \pm 3/2$). This H -grade preference, if present, would propagate through the spectral sum to produce the bias (29). In standard QCD the Atiyah–Singer index theorem fixes the γ_5 chirality of zero modes in nontrivial topological sectors but does not constrain any analog of the H -grade content. No standard QCD argument predicts a preference between the $H = \pm 1/2$ and $H = \pm 3/2$ eigenspaces of the Cl(6) Cartan hypercharge, and we are not aware of any prior work that addresses this question. We offer the hypothesis as a physically motivated guess, with the explicit acknowledgment that the opposite sign (singlet enhanced) and the null result ($\delta f_{\text{triplet}} = 0$) are equally consistent with present knowledge.

We stress that (29) is a hypothesis, not a derived result. Both the magnitude and the sign of $\delta f_{\text{triplet}}$ require the Experiment B data to confirm or refute. The free-field theorem provides the null hypothesis: $\delta f_{\text{triplet}} = 0$. Experiment A establishes this null at heavy mass (0.6σ); Experiment B will test it at the chiral point.

6.8.2 Experiment B: design and falsification criteria

Experiment B will run at $\kappa \approx \kappa_c^{\text{ren}}(\beta)$ determined from Block 1: $\kappa \approx 0.143\text{--}0.145$ at $\beta = 4.5$ and $\kappa \approx 0.150$ at $\beta = 6.0$ (systematic uncertainty ± 0.005 from the $N = 4/N = 6$ spread; see Sec. 5.1). The target observable is the C-even phase difference $\delta f_{\text{triplet}}$ defined in Eq. (29). The three practical complications identified in Sec. 3.3 require specific mitigations before running: (i) CG ill-conditioning near κ_c^{ren} is addressed by deflation of the ten lowest eigenmodes, even–odd Schur complement preconditioning, and adaptive solver tolerance set to 10^{-10} , as detailed in Appendix D; (ii) signal-to-noise degradation at large t restricts the analysis to timeslices satisfying $C_\pi(t)/\sigma_{C_\pi}(t) > 3$ in both phases; (iii) the κ_c^{ren} systematic is controlled by running at two values $\kappa_c^{\text{ren}} \pm 0.003$ and verifying that $\delta f_{\text{triplet}}$ is stable across this range.

The falsification criteria are as follows. A positive outcome— $\delta f_{\text{triplet}} > 0$ at $\geq 3\sigma$ significance with $N_{\text{cfg}} \gtrsim 100$ at $N = 6$ —would constitute evidence for the hypothesis: that the chiral gauge vacuum distributes spinor weight preferentially into the color-carrying channels in the confined phase. Confirmation, in the lattice-QCD sense of an effect that survives systematic checks, would require independent verification by volume scaling to $N = 8$ and a continuum-limit study at multiple β (Sec. 6.8.3); the present work cannot make a confirmation claim on a single volume at a single lattice spacing regardless of per-point significance. A null outcome— $\delta f_{\text{triplet}}$ consistent with zero at $N_{\text{cfg}} \gtrsim 100$ —would establish that grade fractions are insensitive to the confinement–deconfinement transition even at the chiral point, and that the free-field theorem extends to the long-range propagator regime; this result would itself be clean and publishable, establishing the grade fractions as a renormalization-free null observable. An anti-screening outcome— $\delta f_{\text{triplet}} < 0$ —would be physically unexpected and would prompt investigation of whether the effect is genuine (requiring volume scaling) or numerical (CG convergence bias near κ_c^{ren}). Persistence of A_{color} or A_{singlet} beyond the $N_{\text{cfg}}^{-1/2}$ scaling at large statistics would indicate a C-symmetry violation in the code or ensemble update and must be diagnosed before any physics interpretation is attempted.

6.8.3 Diagnostic roadmap

Before any physical interpretation of Experiment B results, a free-field prerequisite test must be performed. With all gauge links set to $U_\mu(x) = I_3$, the projector code, the trace evaluation, and the propagator pipeline must reproduce $f_r = \dim(r)/8$ to machine precision on every configuration, in accordance with Theorem 6.1. This requires no gauge generation and runs in minutes on a single workstation. It is a prerequisite for trusting any nonzero result from the chiral-regime calculation: the test was verified for the heavy-mass production code (Sec. 6.6, where $\delta_{\text{max}} = 5.4 \times 10^{-16}$ was observed) and must be re-verified after any code modifications introduced for the solver mitigations of Sec. 6.8.2. Failure indicates a bug in the projector or trace pipeline and invalidates any subsequent physics interpretation; passing is necessary but not sufficient for the chiral-regime results to be physical.

Beyond this prerequisite, the three mechanisms identified in Sec. 6.7.3 can be

disentangled by a sequence of targeted calculations ordered by computational cost.

The first physics test is N_{cfg} scaling: increasing the ensemble size from $N_{\text{cfg}} = 20$ to $N_{\text{cfg}} \approx 200$ at the current heavy-mass parameters ($\kappa = 0.12346$, $\beta \in \{4.5, 6.0\}$, $N = 6$). If per-channel deviations from $\dim(r)/8$ scale as $N_{\text{cfg}}^{-1/2}$ and approach zero, mechanism (a) is confirmed as the primary driver and no floor from (b) or (c) is present. If deviations approach a nonzero floor, the sign and structure would distinguish a lattice artifact (mechanism b, positive-definite) from a vacuum-dressing signal (mechanism c, phase-dependent).

Separating mechanisms (b) and (c) rigorously requires a continuum-limit study: repeating Experiment B at multiple values of β while holding the physical pion mass fixed by adjusting κ accordingly. Mechanism (b) contributes at $O(a)$ and vanishes as $\beta \rightarrow \infty$; mechanism (c) is physical and survives the continuum limit. A practical program would use $\beta \in \{4.5, 5.5, 6.0, 6.5\}$ with corresponding κ_c^{ren} values from Block 1 (Sec. 5.1), on $N = 6$ or $N = 8$ lattices. Concretely, this requires three extensions beyond the present work: Block 1 must be completed to all six β values; the Wilson action should be improved to remove leading $O(a)$ artifacts; and ensembles must be generated at multiple lattice spacings. Collectively this is a project at least an order of magnitude larger than the present work and is beyond the single-workstation budget within which this program currently operates.

6.8.4 Future directions

Two directions extend the grade-decomposition program beyond what Sec. 6.8.3 addresses. Both lie outside the scope of the present work but are noted as natural targets for follow-up investigation.

The first is the extension to dynamical fermions. In the quenched approximation used throughout, the fermion-loop contributions carrying the $U(1)_A$ anomaly are suppressed by construction; the singlet subspaces $\mathbf{1}_a$ and $\mathbf{1}_b$ are therefore insensitive to topological charge at the present level of approximation. Unquenching could couple the axial anomaly to the grade channels through near-zero Dirac modes, but Proposition 6.2 sharply limits where such an effect can appear. The Spin(6)-chirality combination $A_{\Gamma_7} = A_{\text{color}} + A_{\text{singlet}}$ vanishes identically on every configuration because e_5, e_6 never enter D_W ; unquenching modifies the gauge *measure* but not this operator content, so the identity—and with it the per-configuration lock $A_{\text{singlet}} = -A_{\text{color}}$ (equivalently $f_{1_a} = \frac{1}{2} - f_3$, $f_{1_b} = \frac{1}{2} - f_{\bar{3}}$)—survives unquenching unchanged. The singlet asymmetry therefore cannot acquire a topological-sector dependence *independent* of the color asymmetry: any topological response of A_{singlet} is exactly minus that of A_{color} , and the chirality direction as a whole carries none. The only grade observable in which dynamical-fermion or anomaly effects could register is the within- Γ_7 H -splitting $f_{\text{triplet}} = f_3 + f_{\bar{3}}$ (Sec. 6.7.5)—equivalently the complementary singlet total $f_{1_a} + f_{1_b} = 1 - f_{\text{triplet}}$. Whether the η' anomaly leaves a measurable imprint there is a speculative connection requiring dynamical-fermion ensembles and explicit topological-charge measurement to test.

The second is the connection to geometric-algebra-equivariant machine learning. The four grade-projected correlators $\{C_{\pi}^{(r)}(t)\}$ share properties that make them a

natural target for GA-equivariant neural network architectures: they are gauge-invariant by construction ($[T_a, P_r] = 0$), mutually orthogonal ($P_r P_s = 0$ for $r \neq s$), and labeled by a discrete quantum number—the (Γ_7, H) pair—encoding the $SU(3)$ representation content of each channel. Clifford-equivariant networks [8–11] are designed to respect exactly this type of graded algebraic structure as an inductive bias, and the double grading of Sec. 6.7.2 provides a ready-made label set—triplet, antitriplet, singlet- a , singlet- b —for supervised or unsupervised decomposition of lattice gauge data. Whether this algebraic compatibility translates into practical advantage in generalization or data efficiency remains an open question that the present work does not pursue; we note the connection as a possible application of the $Cl(6)$ framework to geometric-algebra-equivariant deep learning for physical systems, and defer any concrete claims to future investigation.

Finally, the broader programme deferred in Sec. 1.3 remains open as independent follow-up: a renormalized Gell-Mann–Oakes–Renner analysis with Symanzik-improved fermions to control the additive Wilson renormalization that obscures it here, and a direct measurement of the infrared spectral density $\rho(0)$ in the chiral regime to replace the heuristic Banks–Casher argument of Sec. 6.8.1 with a computed quantity. A second, independent determination of κ_c^{ren} —for instance via the m_π^2 extrapolation that proved unstable at $N \leq 6$ (Sec. 5.1)—would also supply the cross-check that the present single-method value lacks.

7 Conclusion

We have constructed and numerically implemented the grade decomposition of the pion correlator in the Clifford algebra $Cl(6)$ framework. The principal findings are as follows.

Algebraic construction. The four-channel decomposition $C_\pi(t) = \sum_r C_\pi^{(r)}(t)$ follows from the Lagrange spectral projectors of the Cartan hypercharge $H = B_{12} + B_{34} + B_{56}$ of $\text{spin}(6)$. The projectors satisfy six independent algebraic identities—completeness, idempotency, orthogonality, $SU(3)$ covariance, charge-conjugation pairing, and γ_5 -commutativity—all verified to machine precision $\leq 2.4 \times 10^{-15}$. The (Γ_7, H) double grading is the standard $SU(4) \supset SU(3) \times U(1)$ Levi decomposition; the present paper is its first dynamical implementation.

Kinematic limit theorem. In the free Wilson–Dirac theory, the grade fractions $f_r(t) = \dim(r)/8$ exactly on every gauge configuration. The proof proceeds through the $I_4 \otimes U(2)$ symmetry of the free propagator, Schur’s lemma, free-field color triviality, and the spatial-cubic invariance of the spatially-summed correlator—the residual time-direction term being removed by its vanishing grade-basis diagonal (Theorem 6.1). The theorem is verified at $\delta_{\text{max}} = 1.26 \times 10^{-16}$ —machine precision—on the free lattice.

Experiment A: heavy-mass production. On thermalized quenched ensembles at $N = 6$, $m_0 = 0.05$, $\beta \in \{4.5, 6.0\}$, $N_{\text{cfg}} = 20$: the algebraic completeness identity holds to $\delta_{\text{max}} = 5.4 \times 10^{-16}$ on every configuration; the measured grade fractions are consistent with the kinematic free-field values at the 10^{-4} level; and the phase differences are consistent with zero at 0.6σ . These results establish that the grade

decomposition machinery works at machine precision and that the heavy-mass result is consistent with the kinematic theorem. They do not address the chiral regime.

Scope statement. Three limitations apply: single lattice volume ($N = 6$), single lattice spacing, and quenched approximation. The central physical question—whether non-perturbative gauge dynamics at the chiral point distinguish the representation channels—requires Experiment B (Sec. 6.8.2), which is identified as the essential follow-up.

Structural observation. The grade decomposition has no natural analog in the standard four-component Wilson–Dirac formulation: the bivector B_{56} (28) requires the internal generators e_5, e_6 of Cl(6) that are absent from Cl(4), and the grade structure cuts across the γ_5 chirality decomposition. The observables $C_\pi^{(r)}(t)$ are accordingly Cl(6)-specific; their gauge invariance, completeness, and SU(3) representation-theoretic interpretation all follow automatically from the algebraic structure without separate proof. A sharper structural fact further constrains what they can measure: because e_5, e_6 do not enter D_W , the Spin(6)-chirality fractions are frozen at $f_{(\Gamma_7=\pm 1)} = \frac{1}{2}$ on every configuration (Proposition 6.2), so the only dynamically resolvable content resides in the within- Γ_7 H -splitting $f_{\text{triplet}} = f_3 + f_{\bar{3}}$ —the observable Experiment B is designed to probe.

Paper 1 [[P1]] established the Cl(6) formulation as a valid and computationally tractable framework for Wilson lattice QCD. The present paper establishes the algebraic infrastructure for grade-decomposed observables and confirms the kinematic null result at heavy mass. Experiment B will determine whether the grade decomposition is a Cl(6)-specific renaming of the standard result or carries genuinely new physical information.

Data and code availability. All simulation code, raw data, and analysis scripts are publicly available under the MIT license. Environment: Python 3.11.9, NumPy 2.2.5, SciPy ≥ 1.11 .

A Cl(6) Basis and Grade Structure

The Cl(6) basis and multiplication structure are presented in full in Paper 1 [[P1]] Appendix A. Here we collect only the elements most directly relevant for the grade projector construction of Sec. 4.

The 15 bivectors $B_{ij} = (i/4)[e_i, e_j]$ are all Hermitian and traceless, with eigenvalues exactly $\{-\frac{1}{2} (\times 4), +\frac{1}{2} (\times 4)\}$. The Cartan hypercharge of the $\text{SU}(4) \supset \text{SU}(3) \times \text{U}(1)$ Levi decomposition is $H = B_{12} + B_{34} + B_{56}$. In the tensor product representation (2),

$$H = -\frac{1}{2}(\sigma_z \otimes I_2 \otimes I_2) - \frac{1}{2}(I_2 \otimes \sigma_z \otimes I_2) - \frac{1}{2}(I_2 \otimes I_2 \otimes \sigma_z),$$

which is diagonal with entries $(-\frac{3}{2}, -\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2})$ in tensor product order, consistent with (8) and the (Γ_7, H) double grading of Sec. 2.4.

The pseudoscalar $I_6 = e_1 e_2 e_3 e_4 e_5 e_6$ satisfies $I_6^2 = -1$, is anti-Hermitian, and anticommutes with all odd-grade elements of Cl(6). The Spin(6) chirality $\Gamma_7 = iI_6$ is Hermitian with eigenvalues ± 1 and commutes with H and with every projector P_r .

The 4D chiral operator $\gamma_5 = e_1 e_2 e_3 e_4$ satisfies $\gamma_5^2 = I_8$, anticommutes with $e_1, \dots, e_4$, and commutes with e_5, e_6, H , and all P_r (verified to 0.00×10^0 in Sec. 4.3). The distinction $\Gamma_7 \neq \gamma_5$ is central to the physics interpretation of Sec. 6.7.5.

B Numerical Verification Details

Environment: Python 3.11.9 · NumPy 2.2.5 · SciPy ≥ 1.11 · IEEE 754 double precision (machine $\varepsilon \approx 2.22 \times 10^{-16}$)

The foundational Cl(6) and SU(3) algebraic verifications are reprised from Paper 1 [[P1]] Appendix B without change. Table B1 of that appendix records the Cl(6) gamma matrix properties; Table B2 the SU(3) generator properties (maximum commutator error 1.24×10^{-16}); Table B5 the coupling structure. Full details are given in Paper 1 [[P1]] Appendix B.

New algebraic verifications introduced in this paper are collected in the table below. All residuals are maximum Frobenius norms over the relevant index set. The final block verifies that the bivector SU(3) octet, derived as the stabilizer of the preferred spinor $|\xi_0\rangle$ (the $H = -\frac{3}{2}$ eigenstate) within the 15-dimensional bivector space, closes as a genuine nonabelian Lie subalgebra of Spin(6), and that its centralizer in $\text{span}(B_{ij})$ is exactly $\text{span}\{H\}$.

Check	Target	Result
H eigenvalue spectrum: $\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\}$, multiplicities $\{1, 3, 3, 1\}$	Exact	✓ Exact
$[\Gamma_7, H] = 0$	0.00×10^0	✓ 0.00×10^0
$[H, T_a] = 0$, all $a \in \{1, \dots, 8\}$	$< 10^{-15}$	✓ $< 10^{-15}$
γ_5 commutativity: $[\gamma_5, P_r] = 0$, all r	0.00×10^0	✓ 0.00×10^0
Charge conjugation: $CP_3C^\dagger = P_3$, $CP_{1_a}C^\dagger = P_{1_b}$	0.00×10^0	✓ 0.00×10^0
$I_6P_{1_a} = -iP_{1_a}$, $I_6P_{1_b} = +iP_{1_b}$	0.00×10^0	✓ 0.00×10^0
<i>Bivector SU(3) octet (stabilizer of $\xi_0\rangle$, $H = -\frac{3}{2}$):</i>		
Stabilizer dimension in $\text{span}(B_{ij})$	8	✓ 8
Lie-algebra closure: $\max \ [X_a, X_b] - \text{proj}_{\text{SU}(3)}[X_a, X_b]\ _F$	$< 10^{-14}$	✓ 6.2×10^{-16}
Nonabelian: $\max \ [X_a, X_b]\ _F$	> 0	✓ 2.0
Centralizer of SU(3) in $\text{span}(B_{ij})$: $\dim = 1$, $ \langle \hat{v}, \hat{H} \rangle $	1.0	✓ 1.0000000000

Checkerboard sweep correctness. The corrected even-odd Metropolis sweep produces equilibrium plaquettes within 0.7% of the Paper 1 [[P1]] reference values at $N = 6$ (Table 2), consistent with the residual difference being attributable to the parallel-sweep systematic in earlier intermediate runs; see Sec. 3.3 for the methodological correction.

C Grade Projector Verification

Table C.1 reports the six projector verification checks referenced in Secs. 4.1 and 4.5. All checks were performed in IEEE 754 double precision, Python 3.11.9, NumPy 2.2.5. Checks C1–C5 must pass to machine precision before any production computation proceeds; they are algebraic prerequisites, not optional verifications.

Table C.1: Grade projector algebraic verification.

Check	Identity	Target	Result	Status
C1	Completeness: $\max \left\ \sum_r P_r - I_8 \right\ _F$	$< 10^{-14}$	0.00×10^0	✓ PASS
C2	Idempotency: $\max \ P_r^2 - P_r\ _F$	$< 10^{-14}$	0.00×10^0	✓ PASS
C3	Orthogonality: $\max \ P_r P_s\ _F, r \neq s$	$< 10^{-14}$	0.00×10^0	✓ PASS
C4	Dimensions: $\text{Tr}(P_3) = 3, \text{Tr}(P_{\bar{3}}) = 3, \text{Tr}(P_{1_a}) = 1, \text{Tr}(P_{1_b}) = 1$	exact integers	exact integers	✓ PASS
C5	SU(3) covariance: $\max \ UP_r U^\dagger - P_r\ _F$ over 50 random $U \in \text{SU}(3)$, all r	$< 10^{-12}$	2.44×10^{-15}	✓ PASS
S1 [†]	Haar average: $\langle \rho(g) \rangle \rightarrow P_{1_a} + P_{1_b}$; $N_{\text{samples}} = 50000$ random $U \in \text{SU}(3)$	$\sim N_{\text{samples}}^{-1/2}$	$0.0050(1/\sqrt{N} \text{ OK})$	✓ PASS

[†] Stochastic check. The algebraic identity is established by C1–C5; S1 verifies behavior under finite-sample group averaging. Expected error $\sigma_{\text{max}}/\sqrt{N} \approx 0.0026$; observed 0.0050; the ratio 1.9 is of the order of, and bounded by, the max-over-element estimate $\sqrt{2 \ln 64} \approx 2.9$. The $1/\sqrt{N}$ scaling is verified across $N_{\text{samples}} = 100$ to 200,000.

D CG Solver Mitigations for the Chiral Regime

The heavy-mass production runs ($m_0 = 0.05$, $\kappa = 0.12346$) required no special solver treatment: mean CG iteration counts were 18.95 at $\beta = 4.5$ and 21.85 at $\beta = 6.0$, with zero solver failures across all $N_{\text{cfg}} = 20$ configurations. This appendix provides the prerequisite recipe for Experiment B (Sec. 6.8.2) at $\kappa \approx \kappa_c^{\text{ren}}$, where the CG condition number $\kappa_{\text{cond}} \sim 1/m_{\text{phys}}^2$ diverges. The three mitigations below are recommended in order of implementation priority.

(i) Deflation of low modes. Compute the 10 lowest eigenmodes of $D_W^\dagger D_W$ using LOBPCG or a Lanczos routine before the main CG sweeps. Subtract the deflated contribution exactly before applying CG; this removes the conditioning bottleneck from the near-zero sector. Expected speedup: 5–10 $\times$ near κ_c^{ren} on $N = 6$ lattices.

(ii) Even-odd Schur complement preconditioning. Decompose the lattice into even and odd sites. The Schur complement $D_{ee} - D_{eo} D_{oo}^{-1} D_{oe}$ has a better condition number than D_W itself at strong coupling. D_{oo} is diagonal in the hopping parameter

κ representation and invertible exactly. This reduces the effective system size by a factor of two and removes the odd-site contribution to conditioning.

(iii) Adaptive tolerance. Within 5% of κ_c^{ren} (estimated from Table 3), tighten the CG stopping criterion to $\text{rtol} = 10^{-10}$ and raise `max_iter` to 5000. Monitor the residual $\|D_W x - b\|/\|b\|$ post-solve; if it exceeds 10^{-3} , discard the configuration and log the failure rather than proceeding with a spurious propagator.

If CG still fails to converge on $N = 10$ lattices after applying all three mitigations, this is a hard limitation to be reported honestly. No results should be produced by forcing convergence. Experiment B is not required to succeed on $N = 10$ for the null-or-signal outcome to be interpretable on $N = 6$.

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